



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

SECI1143 - 11 PROBABILITY & STATISTICAL DATA ANALYSIS

GROUP (Heng Ong Huat) - PROJECT 2

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SECI1143: Probability & Statistical Data Analysis

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PROJECT 2 (10%): Assessment Rubric

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-EVALUATIONS-

Part A: Statistical Analysis: Overall Execution				
Item	Mark			
Hypothesis statement	1	2	3	4
Execution of test	1	2	3	4
Interpretation of results	1	2	3	4
Conclusion and discussion	1	2	3	4
Total	[____/16] * 8 =			
Part B: Statistical Analysis: Selected Test				
Compulsory				
Item	Mark			
1 sample test or 2 sample test	1	2	3	4
Correlation	1	2	3	4
Regression	1	2	3	4
Optional (Choose at least 1)				
Goodness of fit test	1	2	3	4
Chi Square test of independence	1	2	3	4
ANOVA	1	2	3	4
Total	[____/16] * 12 =			

Part C: Format & Content of Report				
Item	Mark			
Introduction	1	2	3	4
Focus on topic (Content)	1	2	3	4
Conclusion (Organization)	1	2	3	4
Grammar & Spelling (Conventions)	1	2	3	4
Total	[____/16] * 10 =			
Part D: Video Presentation				
Item	Mark			
Content Accuracy	1	2	3	4
Sequencing of Information	1	2	3	4
Effectiveness	1	2	3	4
Delivery	1	2	3	4
Videography Clarity	1	2	3	4
Total	[____/20] * 10 =			

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1.0 Introduction

Employers and employees both need to understand the variables that influence spend in the current competitive employment environment. The related dataset from Kaggle that is the focus of the report includes years of experience along with salary information. The dataset was collected from a variety of publicly accessible sources, including job posting sites, surveys, and other public available sources. Five factors were included in the collection of 6699 data points which are age, experience, job role, education level, and salary. When analyzing this kind of data, significant trends and relationships can be found that can help with negotiating salaries, career planning, and organizational salary structuring.

Our main objective in doing this study is to understand the complicated relationship between salary and job experience. Studying these relationships can help people make properly informed choices for their careers and help employers create attractive benefit plans. The objective we have in analyzing this dataset is to provide information on the variables that have an important effect on salary which is a field that has always been of interest in the fields of economics, human resources, and personal finance.

Objectives:

1. To Conduct Statistical Test Analysis:

- The objective is to extract useful information from the dataset by conducting various kinds of statistical tests on data. We will be able to identify important patterns and differences as well as analyze the overall distribution and attributes of the data with the support of these tests.

2. To Determine Dependency of Variables:

- Identifying whether the selected variables such as years of experience and salary, are dependent on one another is a major objective. To determine the course and intensity of their relationship, correlation analysis will be used, along with different statistical methods to test for dependence.

Given the general assumption that people with more work experience generally make greater salaries, we expect to find a beneficial relationship between years of experience and salary. Additionally, we predict that our regression model will provide reliable dependence on experience salary expectations. The alternative analysis may also highlight other relevant correlations and trends in the dataset. By achieving these objectives, this report aims to present an in-depth analysis of the relationship between experiences and salary. The results of the study will be helpful for both people planning their careers and companies trying to set up properly and competitive salary ranges.

2.0 Dataset

Our data set was collected from online sources (Kaggle). This data set consists of 6 different variables and we use all the variables inside the data set, which are age, gender, education level, job title, years of experience, and salary. The table below shows the details of each variable regarding their level of measurement and type of data.

No	Variable / Parameter	Level of Measurement	Type of Data
1.	Age	Ratio	Discrete
2.	Gender	Nominal	Categorical
3.	Education Level	Nominal	Categorical
4.	Job Title	Nominal	Categorical
5.	Years of Experience	Ratio	Discrete
6.	Salary	Ratio	Continuous

Table 1: Type of variables

We chose these variables because all of them are the most common factors that will influence the salary of a person. Thus we want to evaluate whether these variables affect salary or not.

Hypothesis testing, correlation, regression, chi-square test of independence, and ANOVA are used to test the sample data. The ways to do the analysis of data are by using RStudio to generate a graphical presentation of data and do some of the basic calculations. Then, the conclusions are drawn.

Selected Variables	Reason	Test Analysis and Possible Outcome
Gender, Salary	To determine whether gender will affect someone's salary or not	Analysis: 2 Sample Hypothesis Testing Possible Outcome: By conducting a two-sample t-test, we can compare the mean for the male and female. If the P-value is smaller than the significant level (0.05), $p < 0.05$, it would seem that there is a significant difference in salary between male and female
Age, Salary	To examine the relationship between age and salary	Analysis: Correlation Analysis Possible Outcome: By calculating the correlation coefficient between age and salary, we can determine the strength of the relationship between these two variables, age and salary. If the correlation coefficient is positive and statistically significant, we can conclude that as age increases, the salary also increases.
Year of Experience, Salary	To investigate the relationship between the year of experience and salary	Analysis: Regression Analysis Possible Outcome: By conducting a regression analysis, we can determine the relationship between the year of experience and salary. If the regression coefficient is statistically significant (usually $p < 0.05$), it would show that there exists a linear

		relationship between the year of experience and salary. Hence, we can determine the strength of their relationship quantitatively through a regression equation.
Education Level, Salary	Evaluate the independence between education level and salary	Analysis: Chi-Square Test of Independence Possible Outcome: By performing a chi-square test, we can examine whether there is an association between the education level and salary. If the P-value from the chi-square test is significant (usually $p < 0.05$), it indicates that education level and salary are dependent.
Job Title, Salary	To estimate the independence between the job title and salary	Analysis: ANOVA Test Possible Outcome: By performing an ANOVA test, we can analyze the relationship between the job title and salary. If the P-value is smaller than the significant level ($p < 0.05$), we can say that the job title will influence the salary. Thus, the job title and salary are dependent on each other.

Table 2: Statistical Test Analysis

3.0 Data Analysis

3.1 Hypothesis 2 Sample

Based on the dataset, we conduct a 2 sample hypothesis test for this data. The variables selected in this hypothesis test are gender and salary. This hypothesis test is conducted to determine if there is a difference in mean salaries between males and females. The sample size for male is $n_1 = 3674$ and for female is $n_2 = 3014$.

Hypothesis Statement:

- Null hypothesis, $H_0: \mu_1 = \mu_2$
- Alternative hypothesis, $H_1: \mu_1 \neq \mu_2$

Where:

- μ_1 represents the population mean salary for males.
- μ_2 represents the population mean salary for females.

Assume the population variance for the both groups are unequal, then the test statistic formula and the degree of freedom are shows at below:

- Test Statistic Formula:

$$T_0^* = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Where:

$\bar{X}_1$ = sample mean of first sample

$\bar{X}_2$ = sample mean of second sample

s_1^2 = sample variance of first sample

s_2^2 = sample variance of second sample

n_1 = sample size of first sample

n_2 = sample size of second sample

- Degree of Freedom:

$$\nu = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2} \right)^2}{\frac{\left(\frac{S_1^2}{n_1} \right)^2}{n_1 - 1} + \frac{\left(\frac{S_2^2}{n_2} \right)^2}{n_2 - 1}}$$

Test Statistic:

The sample mean of salary for male, $\bar{X}_1 = 121389.9$, while the sample mean of salary for female, $\bar{X}_2 = 107889.0$. A significant level, $\alpha = 0.05$ is used to test the claim that whether there is a difference in mean of salaries between the male and the female. By using RStudio, we obtain the test statistics, $T_0^* = 10.474$ and the degree of freedom, $\nu = 6400.4$. The probability value that we obtain is, $p\text{-value} < 2.2e^{-16}$.

```
Welch Two Sample t-test

data: male_salaries and female_salaries
t = 10.474, df = 6400.4, p-value < 2.2e-16
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 10973.94 16027.81
sample estimates:
mean of x mean of y
 121389.9  107889.0
```

Conclusion:

Since the $p\text{-value} < \alpha$ which is < 0.05 , therefore the null hypothesis, H_0 is rejected. There is sufficient evidence to support the claims that there is a difference in the mean of salary between the male and female. Thus, we can be sure that different genders have different salaries. It is because of the different years of experience for different genders, the amount of salary is different.

3.2 Correlation Test

In the correlation test, we measure the relationship between the age and the salary of each individual in a sample size of 6699. Since both of the data are ratio type, Pearson's technique was selected for calculating the sample correlation coefficient, r . The formula for calculating the sample correlation coefficient, r by using the Pearson technique and the suitable test statistic formula are shown below.

$$\text{Sample Correlation Coefficient, } r = \frac{\Sigma xy - (\Sigma x \Sigma y) / n}{\sqrt{[(\Sigma x^2) - (\Sigma x)^2 / n][(\Sigma y^2) - (\Sigma y)^2 / n]}}$$

$$\text{Test statistic formula, } t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}} \quad \text{degree of freedom, } df = n - 2$$

where,

r = Sample correlation coefficient

n = Sample size

x = Value of the independent variable

y = Value of the dependent variable

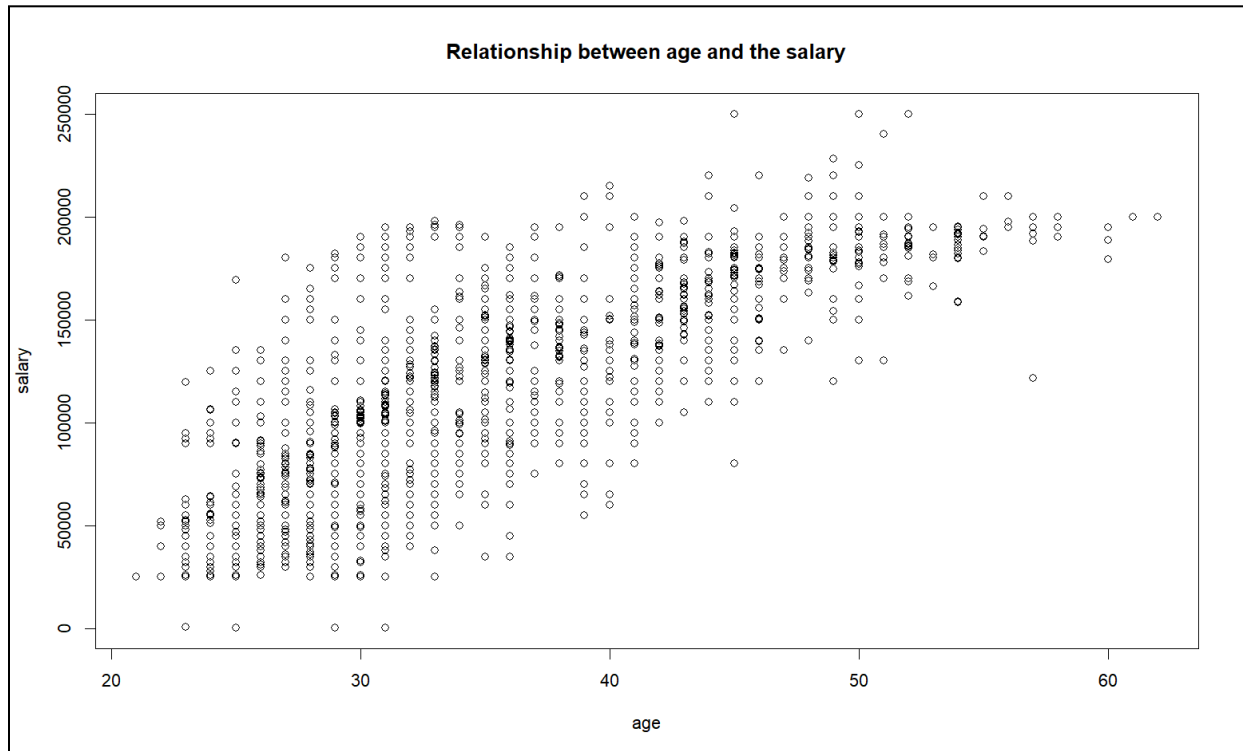


Figure 1: Scatter plot shows the relationship between age and salary

From the scatter plot in Figure 1, it shows that there is a positive linear correlation between age and salary.

Hypothesis statement:

Null Hypothesis, $H_0: \rho = 0$ (no linear correlation)

Alternative hypothesis, $H_1: \rho \neq 0$ (linear correlation exists)

where ρ represents the population correlation coefficient

Test Statistic:

A significant level of 0.05 ($\alpha = 0.05$) is used to test the claim that there exists a linear correlation between the ages and the salary of each individual. From the RStudio we obtain the correlation coefficient, $r = 0.728$, test statistic value, $t = 86.912$ and the degree of freedom is $df = 6697$. The probability value that we obtain is $P\text{-value} = 0$.

Conclusion:

Since the $P\text{-value} < \alpha$ ($0 < 0.05$), thus we reject the null hypothesis, H_0 . There is sufficient evidence to conclude that there exists a linear correlation between age and salary at the 0.05 significance level. As age is positively correlated with salary, this makes common sense, because the larger the age of the people, the more experience in their job scope and their company may prefer to pay more to them. Besides that, the correlation coefficient is height which is 0.728 so there is a strong linear relationship between age and salary.

```
> corTest(dropna_data$Age, dropna_data$Salary)
n = 6699
df = 6697
r = 0.7280526
t value: 86.91246
P-value: 0
Critical t-value at alpha = 0.05 : 1.960318
The correlation is statistically significant at alpha = 0.05
> |
```

3.3 Regression Test

In the regression test, we measure the relationship between the year of experience and salary with a sample size, $n = 6699$. The independent variable, x is the year of experience and the dependent variable, y is salary. The sample regression line provides an estimate of the population regression line.

Estimated Regression Model, $\hat{y}_i = b_0 + b_1x$

where,

$\hat{y}_i$ = estimated dependent variable value

b_0 = estimated of the regression intercept

b_1 = estimated of the regression slope

x = independent variable

Test Statistic Formula, $t = \frac{b_1 - \beta_1}{s_{b_1}}$ degree of freedom, $df = n - 2$

where,

b_1 = Sample regression slope coefficient

β_1 = Hypothesized slope

s_{b_1} = Estimator of the standard error of the slope



Figure 2: Scatter plot relationship between years of experience and salary

From the scatter plot in Figure 2, we can see that there is a positive linear relationship between the year of experience and salary. The best-fit line shows an upward inclining.

From the summary, we can get the formula for the estimated regression model which is

$$Y = 58283.28 + 7046.77X$$

From the formula, the value of $b_0 = 58283.28$ is the estimated value of y when the value of x is equal to 0. While the value of $b_1 = 7046.77$ measures the estimated change in the average value of salary, y as a result of a one-unit change in the year of experience, x.

The coefficient of determination is the portion of the total variation in the dependent variable that is explained by variation in the independent variable.

$$R^2 = \frac{SSR}{SST} = \frac{\text{sum of squares explained by regression}}{\text{total sum of squares}}$$

From the RStudio we obtain the value of the coefficient of determination, $R^2 = 0.6544$, suggesting that approximately 65.44% of the variability in salary can be explained by the years of experience. This indicates a strong linear relationship between years of experience and salary.

Hypothesis statement:

Null hypothesis, $H_0: \beta_0 = 0$ (no linear relationship)

Alternative hypothesis, $H_1: \beta_0 \neq 0$ (linear relationship does exist)

where β represents the population slope coefficient (hypothesized slope).

Test statistic:

A significant level of 0.05 ($\alpha = 0.05$) is used to test the claim that there exists a linear relationship between the year of experience and the salary. From the RStudio we obtain the sample regression slope coefficient, $b_1 = 7046.77$ estimator of the standard error of the slope, $s_{b_1} = 62.57$, test statistic value, $t = 112.62$ and the degree of freedom is $df = 6697$. The probability value that we obtain is the P-value $= 2.2 \times 10^{-16}$.

Conclusion:

Since $P\text{-value} < \alpha$ or $(2.2 \times 10^{-16} < 0.05)$, thus reject null hypothesis, H_0 . There is sufficient evidence to support the claims that there exists a linear relationship between the year of experience and the salary. The residual standard error is 31030, indicating the average amount that observed salaries deviate from the regression line. The R^2 value of 0.6544 suggests a strong linear relationship.

Call:

```
lm(formula = y ~ x)
```

Residuals:

Min	1Q	Median	3Q	Max
-148236	-22377	-5564	21015	100576

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	58283.28	632.71	92.12	<2e-16 ***
x	7046.77	62.57	112.62	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 31030 on 6697 degrees of freedom

Multiple R-squared: 0.6544, Adjusted R-squared: 0.6544

F-statistic: 1.268e+04 on 1 and 6697 DF, p-value: < 2.2e-16

[1] 0.8089689

$$b_0 = 58283.28$$

$$b_1 = 7046.77$$

$$s_{b_1} = 62.57$$

$$s_\varepsilon = 31030$$

$$df = 6697$$

$$R^2 = 0.6544$$

$$p\text{-value} = 2.2 \times 10^{-16}$$

3.4 Chi-Square Test of Independence

One way to determine whether two variables are likely to be related or not is to apply the Chi-square test of independence. In order to proceed with the test, we summarized the data using a two-way contingency table. In this test, we will test whether there is a relationship between different education levels in terms of salary distribution. The formula for the test statistic is:

Test statistic formula, $\chi^2 = \sum_{all\ cells} \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$; Degree of freedom, $df = (r - 1)(c - 1)$

Where,

o_{ij} = Observed count at row i column j

e_{ij} = Expected count at row i column j

r = total number of rows

c = total number of columns

Hypothesis statements:

Null hypothesis, H_0 : salary distributions are independent of a person's education level.

Alternative hypothesis, H_1 : salary distributions are dependent on a person's education level.

Test statistics

A significant level of 0.05 ($\alpha = 0.05$) is used to test the claim that salaries given to a person are dependent on the individual's education level. The statistics below show the two-way contingency table created with the help of RStudio.

Education Level	(349.9995,75244 .99965]	(7.52e+04,1e+05]	(1e+05,1.25e+05]	(125174.99975, 250000]
Bachelor's Degree	1357	591	335	738
High School	428	6	4	10
Master's Degree	189	271	442	958
PhD	38	48	73	1210

Table 3: Observed frequency table

Education Level	(349.9995,75244.99965]	(7.52e+04,1e+05]	(1e+05,1.25e+05]	(125174.99975,250000]
Bachelor's Degree	907.4727	413.14362	385.17976	1315.2039
High School	134.5739	61.26724	57.12033	195.0385
Master's Degree	558.7220	254.36847	237.15139	809.7581
PhD	411.2314	187.22066	174.54852	595.9994

Table 4: Expected frequency table

From the RStudio we obtain the test statistic value, $\chi^2 = 3057.4$ and the degree of freedom is $df = 9$. The probability value that we obtained is $P - value = 2.2 \times 10^{-16}$.

Conclusion

Since $P - value < \alpha$ or $2.2 \times 10^{-16} < 0.050$, thus the null hypothesis, H_0 , is rejected. There is sufficient evidence to support the argument that salary distributions are determined by a person's educational level, whether it be high school, bachelor's degree, master's degree, or PhD. This is because people with greater education levels typically occupy higher positions in the workplace, resulting in better compensation.

3.5 ANOVA Test

The purpose of the ANOVA test is to evaluate the significant differences between means of salary distribution for all 191 different jobs. We use a significance level of 0.05 to test the null hypothesis that the salary distributions for all 191 different jobs will have the same mean. Due to the different sample sizes of each job being compared, one-way ANOVA was applied with the Unequal Sample Sizes technique.

Test statistic formula, $F = \frac{SST(k-1)}{SSE(n-k)}$

Numerator degrees of freedom = $k - 1$

Denominator degrees of freedom = $n - k$

Where,

$$SST = \sum_{i=1}^k n_i (\bar{Y}_i - \bar{Y})^2$$

$$SSE = \sum_{i=1}^k \sum_{j=1}^{n_i} (\bar{Y}_{ij} - \bar{Y}_i)^2$$

$$\bar{Y} = \frac{1}{k} \sum_{i=1}^k \bar{Y}_i$$

k = number of population or treatments being compared

$n = \sum_{i=1}^k n_i$; n_i = sample size for population i

Hypothesis statement

Null hypothesis, $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \dots \mu_{191}$

Alternative hypothesis, $H_1 : \text{at least one mean is different}$

where all the 191 jobs' means are compared.

Test statistic

A significant level of 0.05 ($\alpha = 0.05$) is used to test the claim that the salary distributions for people in different jobs will have the same mean. From the calculations done in RStudio, we obtained the test statistic value $F = 79.743$ and the degree of freedom, $df = 6697$. The probability value we have obtained is $P - \text{value} = 0$.

Conclusion

Since $P - \text{value} < \alpha$ or $0.00 < 0.050$, thus the null hypothesis, H_0 , is rejected. There is sufficient evidence to support the claims that the salary distributions for people in different jobs will have at least one different mean. The salary pay cannot be assumed to be the same for every single job, as some require more skills and experience to work with, while others might have low requirements and be easier to work around.

4.0 Conclusion

In this project, we participated in several phases of data analysis, starting with choosing a suitable dataset before proceeding through data pre-processing and different statistical method analysis. During the dataset selection process, a Kaggle dataset with a focus on salary data was chosen. This dataset includes variables like age, gender, job title, years of experience, education level, and salary. This dataset provided for analyzing the factors affecting salaries and allowed for a variety of statistical tests. We determined the data that are missing, to ensure we could test for accuracy in the data, and processed the data properly for analysis during the pre-processing stage by using RStudio. It is used to ensure that the analysis that proceeded generated reliable results, this stage was needed to check the null data in our dataset.

In the analysis process phase, we conducted several statistical tests, including hypothesis test, correlation test, regression test, chi-square test of independence and ANOVA test by using RStudio. These tests provided different findings on how the variables influence the salaries. In the hypothesis test, we reject the null hypothesis since the p-value is smaller than the significant level. This result shows that there could be differences in salaries between each gender. After processing the correlation test, we reject the null hypothesis as well. The conclusion is made as there exists a linear correlation between age and salary at the 0.05 significance level. For the regression test, we reject the null hypothesis. We claim that there exists a linear relationship between the year of experience and the salary as the residual and residual standard error which is the difference between y and predicted y is larger, r square is low. Thus, there is a weak linear relationship. For the chi-square test of independence, the null hypothesis is rejected. It is because the salary distributions are determined by a person's educational level, whether it be high school, bachelor's degree, master's degree, or PhD. It means that people with greater education levels typically occupy higher positions in the workplace, resulting in better compensation. Through the ANOVA test, we reject the null hypothesis. We claim that the salary distributions for people in different jobs will make at least one different mean. It is inaccurate to assume that all jobs reward equally because certain jobs may have lower requirements and be less complicated to work around, whereas some might need more experiences and skills.

In conclusion, this project's analysis provided readers an in-depth understanding of the variables influencing salary. The results point to gender-based salaries gaps, an effective connection between age and salary, and show that years of experience have significant effects on salary. Employers and employees can use these statistics to clarify salary variations and make responsible decisions about salary planning and career planning. Through careful choosing of the dataset, pre-processing, and the use of many statistical methods, significant relationships and trends were found within the data. The result shows the importance that proper data analysis is to extract beneficial conclusions from datasets.

5.0 Appendix

Sample of Raw Dataset

Age	Gender	Education Level	Job Title	Years of Experience	Salary
32	Male	Bachelor's	Software Engineer	5	90000
28	Female	Master's	Data Analyst	3	65000
45	Male	PhD	Senior Manager	15	150000
36	Female	Bachelor's	Sales Associate	7	60000
52	Male	Master's	Director	20	200000
29	Male	Bachelor's	Marketing Analyst	2	55000
42	Female	Master's	Product Manager	12	120000
31	Male	Bachelor's	Sales Manager	4	80000
26	Female	Bachelor's	Marketing Coordinator	1	45000
38	Male	PhD	Senior Scientist	10	110000
29	Male	Master's	Software Developer	3	75000
48	Female	Bachelor's	HR Manager	18	140000
35	Male	Bachelor's	Financial Analyst	6	65000
40	Female	Master's	Project Manager	14	130000
27	Male	Bachelor's	Customer Service Rep	2	40000
44	Male	Bachelor's	Operations Manager	16	125000
33	Female	Master's	Marketing Manager	7	90000
39	Male	PhD	Senior Engineer	12	115000
25	Female	Bachelor's	Data Entry Clerk	0	35000
51	Male	Bachelor's	Sales Director	22	180000
34	Female	Master's	Business Analyst	5	80000
47	Male	Master's	VP of Operations	19	190000
30	Male	Bachelor's	IT Support	2	50000
36	Female	Bachelor's	Recruiter	9	60000
41	Male	Master's	Financial Manager	13	140000

Data Resource Link:

<https://www.kaggle.com/datasets/mohithsairamreddy/salary-data/data>

Video Presentation Link:

<https://youtu.be/FaZO2N8c4QE>

E-portfolio:

[LOH HUI YI](#)

[LEE JIAN AI](#)

[ONG JIE MIN](#)

[JUSTIN FAUZADANI](#)